```
# Install required packages (Colab usually has them, but safe to run)
!pip install pandas numpy matplotlib scikit-learn

# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

from mpl_toolkits.mplot3d import Axes3D
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0
```

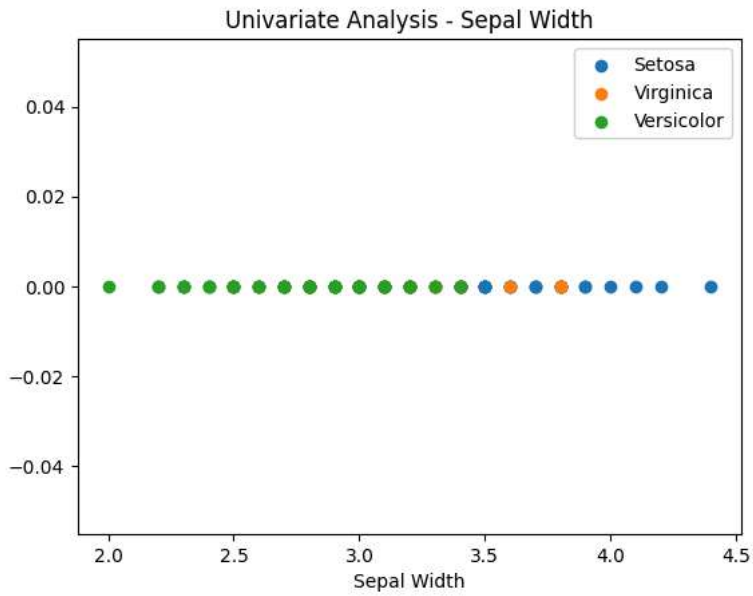
```
# Import libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

# Load dataset (Colab - seaborn built-in dataset)
df = sns.load_dataset('iris')

# Display data
print(df.head())
print("Shape:", df.shape)
```

```
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa
Shape: (150, 5)
```

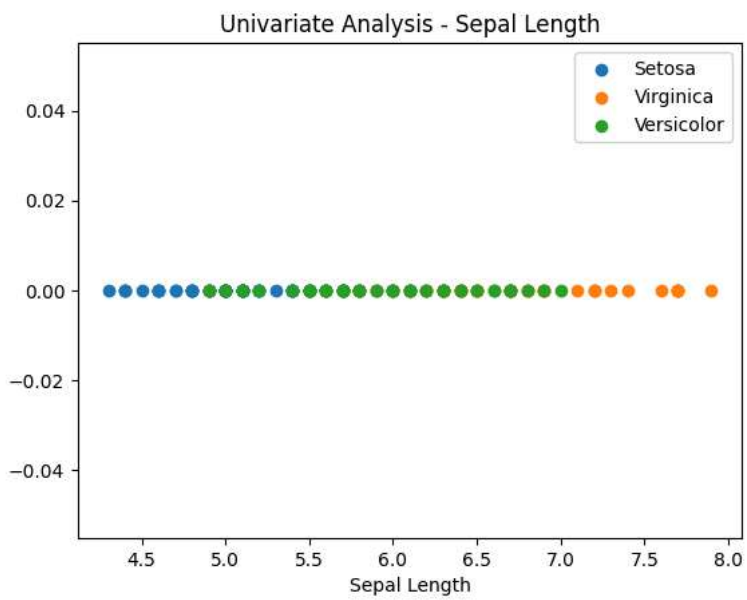
Double-click (or enter) to edit

```
#UNIVARIATE ANALYSIS
```

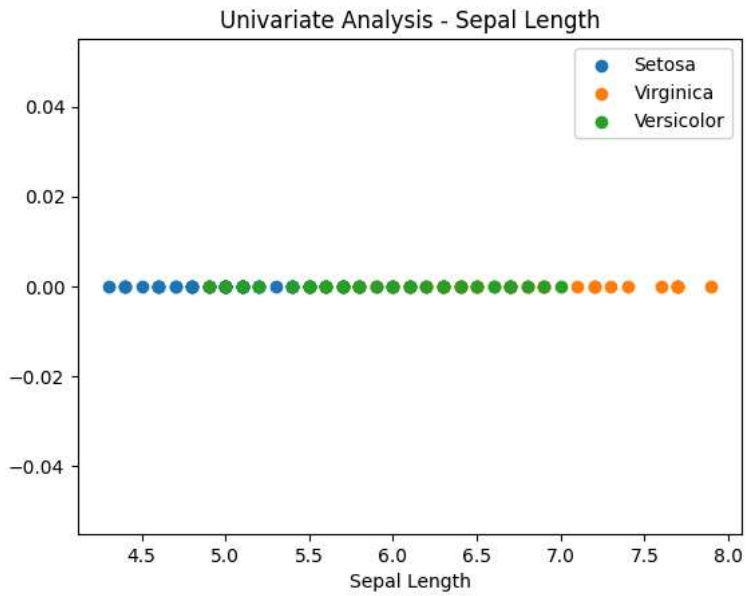
```
df_Setosa = df[df['species'] == 'setosa']
df_Virginica = df[df['species'] == 'virginica']
df_Versicolor = df[df['species'] == 'versicolor']

plt.scatter(df_Setosa['sepal_width'], np.zeros_like(df_Setosa['sepal_width']), label='Setosa')
plt.scatter(df_Virginica['sepal_width'], np.zeros_like(df_Virginica['sepal_width']), label='Virginica')
plt.scatter(df_Versicolor['sepal_width'], np.zeros_like(df_Versicolor['sepal_width']), label='Versicolor')

plt.xlabel('Sepal Width')
plt.title("Univariate Analysis - Sepal Width")
plt.legend()
plt.show()
```



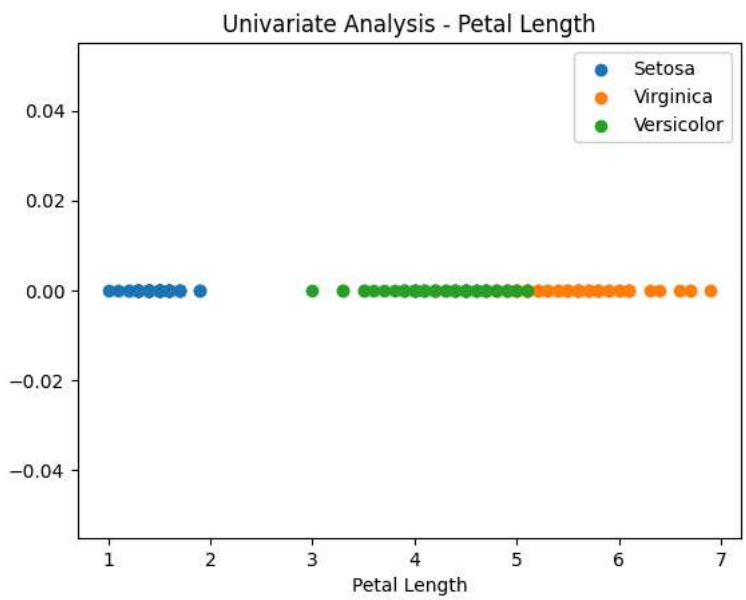

```
plt.scatter(df_Setosa['sepal_length'], np.zeros_like(df_Setosa['sepal_length']), label='Setosa')
plt.scatter(df_Virginica['sepal_length'], np.zeros_like(df_Virginica['sepal_length']), label='Virginica')
plt.scatter(df_Versicolor['sepal_length'], np.zeros_like(df_Versicolor['sepal_length']), label='Versicolor')

plt.xlabel('Sepal Length')
plt.title("Univariate Analysis - Sepal Length")
plt.legend()
plt.show()
```




```
plt.scatter(df_Setosa['sepal_length'], np.zeros_like(df_Setosa['sepal_length']), label='Setosa')
plt.scatter(df_Virginica['sepal_length'], np.zeros_like(df_Virginica['sepal_length']), label='Virginica')
plt.scatter(df_Versicolor['sepal_length'], np.zeros_like(df_Versicolor['sepal_length']), label='Versicolor')

plt.xlabel('Sepal Length')
plt.title("Univariate Analysis - Sepal Length")
plt.legend()
plt.show()
```

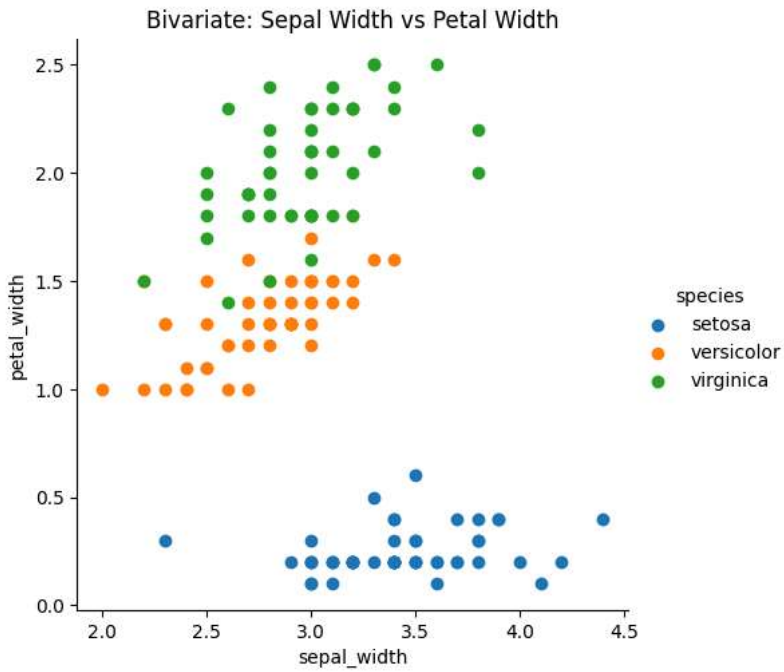



```
plt.scatter(df_Setosa['petal_length'], np.zeros_like(df_Setosa['petal_length']), label='Setosa')
plt.scatter(df_Virginica['petal_length'], np.zeros_like(df_Virginica['petal_length']), label='Virginica')
plt.scatter(df_Versicolor['petal_length'], np.zeros_like(df_Versicolor['petal_length']), label='Versicolor')

plt.xlabel('Petal Length')
plt.title("Univariate Analysis - Petal Length")
plt.legend()
plt.show()
```



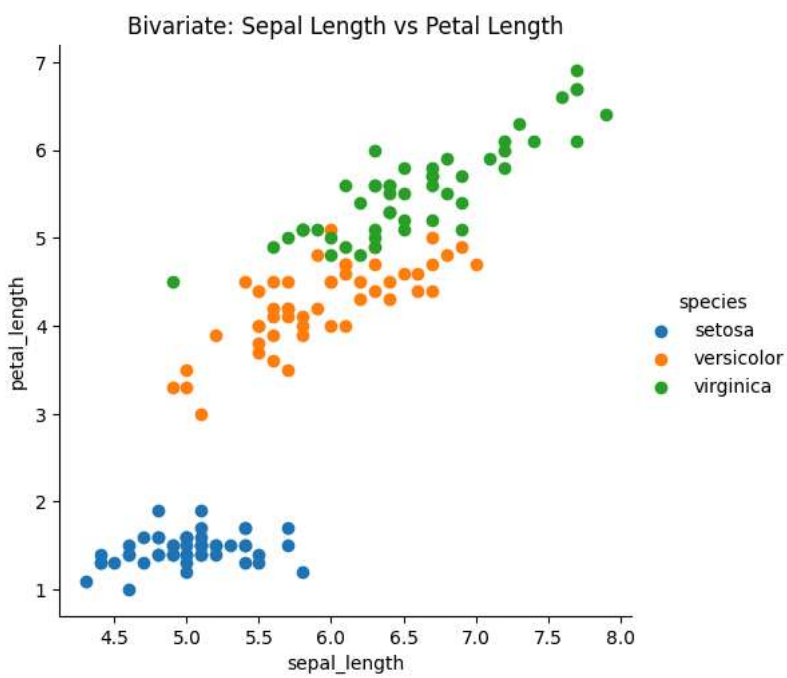

```
#BIVARIATE ANALYSIS
```

```
sns.FacetGrid(df, hue='species', height=5)\
   .map(plt.scatter, "sepal_width", "petal_width")\
   .add_legend()

plt.title("Bivariate: Sepal Width vs Petal Width")
plt.show()
```



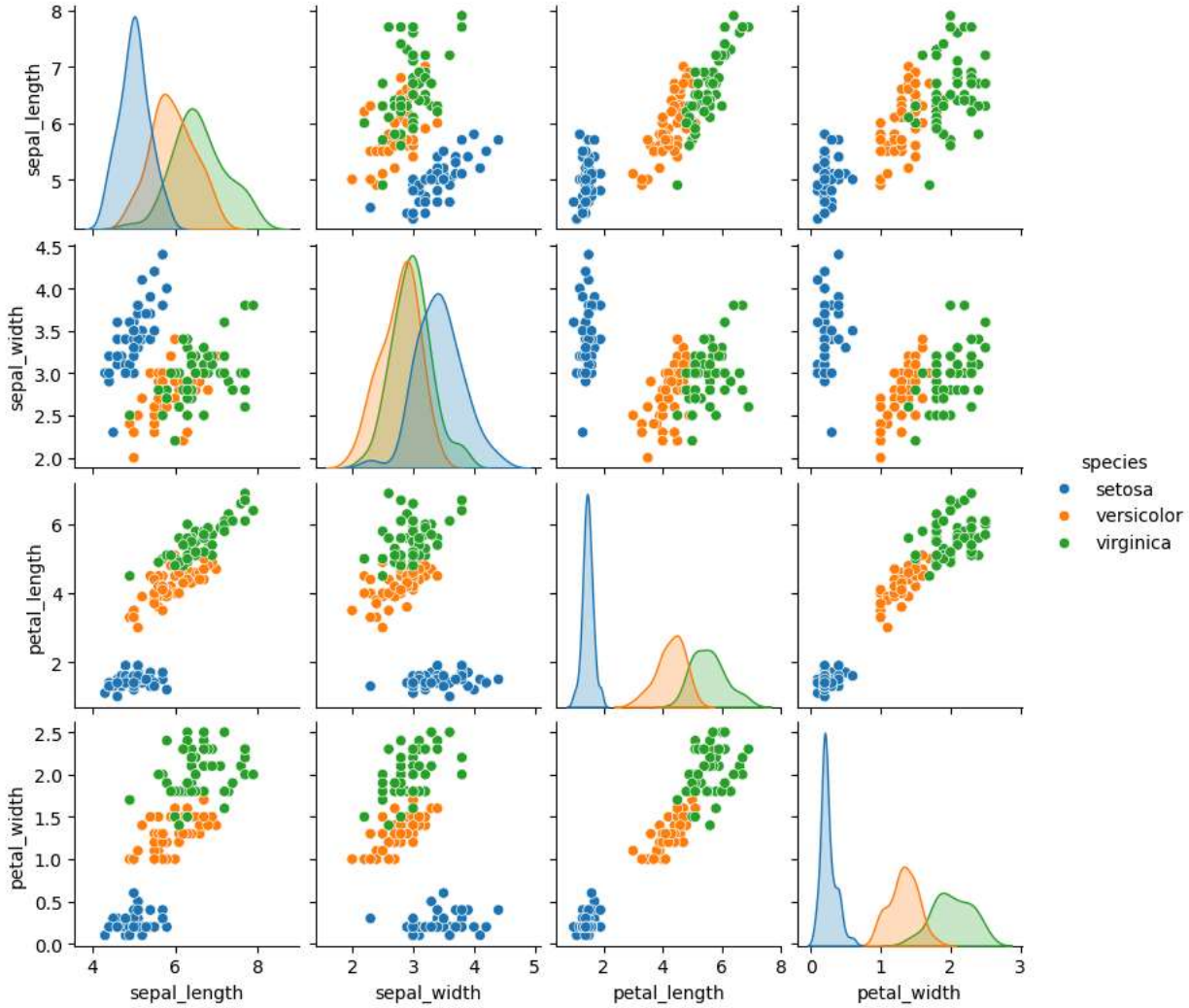
Double-click (or enter) to edit

```
sns.FacetGrid(df, hue='species', height=5)\
   .map(plt.scatter, "sepal_length", "petal_length")\
   .add_legend()

plt.title("Bivariate: Sepal Length vs Petal Length")
plt.show()
```




```
#MULTIVARIATE ANALYSIS
```

```
sns.pairplot(df, hue="species", height=2)
plt.show()
```



Start coding or generate with AI.